TOPIC 2 Transforming and Comparing Linear Functions

Name _____ Date _____

Reminders

Slope = $(y_2 - y_1) \div (x_2 - x_1)$ · $y = mx + b$ · point-slope: $y - y_1 = m(x - x_1)$

Parallel: same slope · Perpendicular: flip the fraction and change the sign (product -1)

Outside f(x) + k: up/down · Inside f(x − h): right h, f(x + h): left h (opposite the sign)

Horizontal: y = #, slope 0 · Vertical: x = #, slope undefined (HOY VUX)

Domain = inputs (x) · Range = outputs (y) · Zero = x-intercept

1. Kai graphed $f(x)$ and $g(x) = f(x + 5)$. Which statement describes how the graphs are related?

A. translated right 5 units

B. translated up 5 units

C. translated left 5 units

D. translated down 5 units

Inside the parentheses the graph moves opposite the sign: + 5 → left 5.

2. A pool rental costs $50 plus $30 per hour, for at most 4 hours. What is the range for one rental?

F. $0 \le x \le 4$

G. $30 \le y \le 170$

H. $50 \le y \le 200$

J. **$50 \le y \le 170$**

Range = costs. 0 hours: $50. 4 hours: 50 + 30(4) = $170.

3. What is the slope of a line perpendicular to $y = -4/5x + 2$?

A. **5/4**

B. −5/4

C. 4/5

D. −4/5

Flip −4/5 to −5/4 and change the sign: 5/4. Check: (−4/5)(5/4) = −1.

4. A line passes through (0, −3) and (6, 0). What is the zero of the function?

F. −3

G. 6

H. 1/2

J. −6

The zero is where y = 0: the point (6, 0). −3 is the y-intercept.